Appendix B
Laplace transform

Control systems tend to have input–output relationships which are described by differential equations. The differential equations describe how the output varies with time for an input. We can determine the output for a system with some particular input by solving the differential equation. However, the Laplace transform is a way of transforming differential equations into more convenient forms for determining outputs.

A quantity which is a function of time can be represented as $f(t)$ and is said to be in the *time domain*, e.g. if we have a voltage v which is a function of time we can write as $v(t)$ to show that it is. In discussing control systems we are only concerned with values of time greater than or equal to 0, i.e. $t \geq 0$, and so to obtain the Laplace transform of $f(t)$ function we multiply it by e^{-st} and then integrate with respect to time from zero to infinity; s is a constant with the unit of 1/time. The result is then said to be in the *s-domain*. The Laplace transform of the function of time $f(t)$, which is written as $\mathcal{L}\{f(t)\}$, is thus:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\, dt$$

A function of s is written as $F(s)$. It is usual to use a capital letter F for the Laplace transform and a lower-case letter f for the time-varying function $f(t)$. Thus:

$$\mathcal{L}\{f(t)\} = F(s)$$

The procedure for using the Laplace transform is to:

1 Transform functions of time into functions of s.

2 Carry out algebraic manipulations in the s-domain. We can carry out algebraic manipulations on a quantity in the s-domain, i.e. adding, subtracting, dividing and multiplying in the way we do with any algebraic quantities.

3 Transform the functions of s back into functions of time, i.e. the so-termed inverse operation. This involves finding the time domain function that could have given the s-domain expression obtained by the algebraic manipulations.

For the inverse operation, when the function of time is obtained from the Laplace transform, we can write

$$f(t) = \mathcal{L}^{-1}\{F(s)\}$$

Obtaining the Laplace transform

The following examples illustrate how we can obtain, from first principles, the Laplace transform of functions of time. However, in practice, engineers use a table of Laplace transforms to avoid having to carry out such calculations; Table 3.1 shows some of the transformations commonly encountered.

The unit step is the type of function encountered when we suddenly apply an input to a system (Figure Ap.B.1). It describes an abrupt change in some quantity from zero to a steady value, e.g. the change in the voltage applied to a circuit when it is suddenly switched on. Thus, at time $t = 0$ it suddenly switches to the value 1 and has the constant value of 1 for all values of time greater than 0, i.e. $f(t) = 1$ for $t \geq 0$. The Laplace transform of the unit step function is:

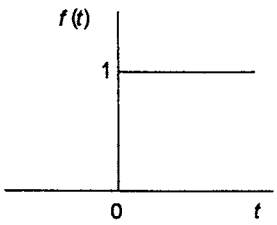

Figure Ap.B.1 *Unit-step function*

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty 1\,e^{-st}\,dt = -\frac{1}{s}[e^{-st}]_0^\infty$$

For $t = \infty$ the value of the exponential term $e^{-\infty}$ is 0 and with $t = 0$ the value of e^{-0} is -1. Hence:

$$F(s) = \frac{1}{s}$$

As a further illustration, consider the Laplace transform of the function $f(t) = e^{at}$, where a is a constant :

$$F(s) = \int_0^\infty e^{at}e^{-st}\,dt$$

$$= \int_0^\infty e^{-(s-a)t}\,dt = -\frac{1}{s-a}\left[e^{-(s-a)t}\right]_0^\infty$$

When $t = \infty$ the term in the brackets becomes 0 and when $t = 0$ it becomes -1. Thus:

$$F(s) = \frac{1}{s-a}$$

Properties of the Laplace transform

To use the Laplace transforms of functions given in tables it is necessary to know the properties of such transforms. The following are basic properties:

1 *Linearity property*
The Laplace transform of the sum of two time functions, e.g. $f(t)$ and $g(t)$, can be obtained by adding the Laplace transforms of the two separate Laplace transforms, i.e.

$$\mathcal{L}\{f(t) + g(t)\} = \mathcal{L}f(t) + \mathcal{L}g(t)$$

Thus the Laplace transform of $1 + t$ is:

$$\mathcal{L}\{1 + t\} = \mathcal{L}\,1 + \mathcal{L}\,t = \frac{1}{s} + \frac{1}{s^2}$$

2 *Multiplication by a constant*
As a consequence of the linearity property, suppose we want the Laplace transform of $2t$. This can be considered to be $t + t$ and so the Laplace transform of $2t$ is the sum of the transforms of t and t and so $2 \times$ the Laplace transform of t. Since the Laplace transform of t is $1/s$ then the Laplace transform of $2t$ is $2/s$. The Laplace transform $4t^2$ is given by 4 times the Laplace transform of t^2 and so, since the Laplace transform of t^2 is $2/s^2$, is $8/s^2$. In general:

$$\mathcal{L}\{af(t)\} = a\mathcal{L}f(t)$$

The Laplace transform of $1 + 2t + 4t^2$ is given by the sum of the transforms of the individual terms in the expression and so is:

$$F(s) = \frac{1}{s} + \frac{2}{s^2} + \frac{8}{s^3}$$

3 *Final-value theorem*
This theorem enables us to determine the value a function of time will end up with after a long period of time, i.e. the steady-state value. The final-value theorem can be stated as: if a function of time $f(t)$ has a Laplace transform $F(s)$ then in the limit as the time tends to infinity the value of the function is given by

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$$

For example, consider the final value of e^{-3t} as t tends to infinity. Using Table 3.1, we have $e^{-at} = 1/(s + a)$ and so $e^{-3t} = 1/(s + 3)$. The final-value theorem thus gives:

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s) = \lim_{s\to\infty} \frac{s}{s+3} = 0$$

and so the steady-state value is 0.

4 *Derivatives*
The Laplace transform of a derivative of a function $f(t)$ is given by:

$$\mathcal{L}\left\{\frac{\mathrm{d}}{\mathrm{d}t} f(t)\right\} = sF(s) - f(0)$$

where $f(0)$ is the value of the function when $t = 0$. For a second derivative:

$$\mathcal{L}\left\{\frac{d^2}{dt^2}f(t)\right\} = s^2F(s) - sf(0) - \frac{d}{dt}f(0)$$

where $df(0)/dt$ is the value of the first derivative at $t = 0$.

For example, if we have a function of time with an initial zero value, i.e. $f(0) = 2$, then the Laplace transform of $df(t)/dt$ is $sF(s)$. If the function had $f(0) = 2$ then the Laplace transform of $df(t)/dt$ is $sF(s) - f(0) = sF(s) - 2$.

As another illustration, if we have $d^2f(t)/dt^2$ with $f(0) = 0$ and $df(0)/dt = 0$ then the Laplace transform of $d^2f(t)/dt^2$ is $s^2F(s)$. If the function had $f(0) = 2$ and $df(0)/dt = 1$ then the Laplace transform of $d^2f(t)/dt^2$ is $s^2F(s) - 2s - 1$.

5 *Integrals*

The Laplace transform of the integral of a function $f(t)$ which has a Laplace transform $F(s)$ is given by:

$$\mathcal{L}\left\{\int_0^t f(t)\,dt\right\} = \frac{1}{s}F(s)$$

For example, the Laplace transform of the integral of the function e^{-t} between the limits 0 and t is:

$$\mathcal{L}\left\{\int_0^t e^{-t}\,dt\right\} = \frac{1}{s}\mathcal{L}\{e^{-t}\} = \frac{1}{s(s+1)}$$

The inverse transform

The inverse Laplace transformation is the conversion of a Laplace transform $F(s)$ into a function of time $f(t)$. This operation is written as:

$$\mathcal{L}^{-1}\{F(s)\} = f(t)$$

The linearity property of Laplace transforms means that if we have a transform as the sum of two separate terms then we can take the inverse of each separately and the sum of the two inverse transforms is the required inverse transform.

$$\mathcal{L}^{-1}\{aF(s) + bG(s)\} = a\mathcal{L}^{-1}F(s) + b\mathcal{L}^{-1}G(s)$$

To illustrate how rearrangement of a function can often put it into the standard form shown in Table 3.1, the inverse transform of $3/(2s + 1)$ can be obtained by rearranging it as:

$$\frac{3(1/2)}{s+(1/2)}$$

Table 3.1 contains the transform $1/(s + a)$ with the inverse of e^{-at}. Thus the inverse transformation is just this transform with $a = ½$ and multiplied by the constant (3/2) and so is $(3/2)\,e^{-t/2}$.

Expressions often have to be put into simpler standard forms of Table 3.1 by the use of partial fractions. For example, the inverse Laplace

transform of $(3s - 1)/s(s - 1)$ is obtained by first simplifying it using partial fractions:

$$\frac{3s-1}{s(s-1)} = \frac{A}{s} + \frac{B}{s-1}$$

and so we must have $3s - 1 = A(s - 1) + Bs$. Equating coefficients of s gives $3 = A + B$ and equating numerical terms gives $-1 = -A$. Hence:

$$\frac{3s-1}{s(s-1)} = \frac{1}{s} + \frac{2}{s-1}$$

The inverse transform of $1/s$ is 1 and the inverse of $1/(s - 1)$ is e^t. Thus:

$$\mathcal{L}^{-1}\left\{\frac{3s-1}{s(s-1)}\right\} = 1 + 2\,e^t$$

Solving differential equations using Laplace transforms

Laplace transforms offer a method of solving differential equations. The procedure adopted is:

1 Replace each term in the differential equation by its Laplace transform, inserting the given initial conditions.

2 Algebraically rearrange the equation to give the transform of the solution.

3 Invert the resulting Laplace transform to obtain the answer as a function of time.

As an illustration, the first-order differential equation:

$$3\frac{dx}{dt} + 2x = 4$$

has the Laplace transform:

$$3[sX(s) - x(0)] + 2X(s) = \frac{4}{s}$$

Given that $x = 0$ at $t = 0$:

$$3sX(s) + 2X(s) = \frac{4}{s}$$

Hence:

$$X(s) = \frac{4}{s(3s+2)}$$

Simplifying by the use of partial fractions:

$$\frac{4}{s(3s+2)} = \frac{A}{s} + \frac{B}{3s+2}$$

Hence $A(3s + 2) + Bs = 4$ and so $A = 2$ and $B = -2/3$. Thus:

$$X(s) = \frac{2}{s} - \frac{2}{3(3s+2)} = \frac{2}{s} - 2\frac{\frac{2}{3}}{\frac{2}{3}\left(s+\frac{2}{3}\right)}$$

and so:

$$x(t) = 2 - 2\,e^{-2t/3}$$

As a further illustration, the second-order differential equation:

$$\frac{d^2x}{dt^2} - 5\frac{dx}{dt} + 6x = 2\,e^{-t}$$

has the Laplace transform:

$$s^2X(s) - sx(0) - \frac{d}{dt}x(0) - 5[sX(s) - x(0)] + 6X(s) = \frac{2}{s+1}$$

Given that $x = 0$ and $dx/dt = 1$ at $t = 0$:

$$s^2X(s) - 1 - 5sX(s) + 6X(s) = \frac{2}{s+1}$$

Hence:

$$X(s) = \frac{\frac{2}{s+1} + 1}{s^2 - 5s + 6} = \frac{2}{(s+1)(s-2)(s-3)} + \frac{1}{(s-2)(s-3)}$$

We can simplify the above expression by the use of partial fractions:

$$\frac{2}{(s+1)(s-2)(s-3)} = \frac{A}{s+1} + \frac{B}{s-2} + \frac{C}{s-3}$$

Hence $A(s - 2)(s - 3) + B(s + 1)(s - 3) + C(s + 1)(s - 2) = 2$ and so $A = 1/6$, $B = -2/3$ and $C = ½$. For the other fraction:

$$\frac{1}{(s-2)(s-3)} = \frac{D}{s-2} + \frac{E}{s-3}$$

Hence $D(s - 3) + E(s - 2) = 1$ and so $D = -1$ and $E = 1$. Thus:

$$X(s) = \frac{\frac{1}{6}}{s+1} + \frac{-\frac{2}{3}}{s-2} + \frac{\frac{1}{2}}{s-3} + \frac{-1}{s-2} + \frac{1}{s-3}$$

$$= \frac{\frac{1}{6}}{s+1} - \frac{\frac{5}{3}}{s-2} + \frac{\frac{3}{2}}{s-3}$$

The inverse transform is thus:

$$x(t) = \tfrac{1}{6}\,e^{-t} - \tfrac{5}{3}\,e^{2t} + \tfrac{3}{2}\,e^{3t}$$